

Observer-Based Sliding Mode Control of a 2-DOF Helicopter System

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Abstract—A sliding mode control (SMC) strategy is presented for a 2 degrees of freedom (DOF) helicopter system, which achieves asymptotic attitude regulation to desired set points as well as trajectory tracking. A rigorous Lyapunov-based analysis is provided to prove convergence to the desired set points. To overcome the practical limitations of velocity measurements, a sliding mode observer is designed to estimate the angular velocities. The convergence of the estimated velocities to the actual velocities is proven via a rigorous Lyapunov-based analysis and illustrated through computer simulations. In addition, computer simulations are presented, to demonstrate the effectiveness of the control law when applied to the complete nonlinear system dynamics. Finally, the control strategy is experimentally tested on the Quanser 2-DOF AERO helicopter. The results of the observer-based SMC strategy are compared to a conventional PID-controller.

I. INTRODUCTION

In the field of robotics, the use of Unmanned Aerial Vehicles (UAVs), more commonly known as drones, has drastically increased over the recent years. One of the most commonly used UAVs is the helicopter system. Helicopters serve a wide range of applications, including: transportation, rescue operations, fire fighting, traffic control, and tourism [1]. Helicopter systems are nonlinear, highly cross-coupled, unstable and difficult to model [2]. Moreover, system parameter identification is a complex problem, a number of experiments are described to determine all system parameters in [3]. Due to the inaccuracy of these experiments the model has to deal with a significant amount of parameter uncertainty.

Helicopter systems are well studied in literature, linear strategies that are employed for the control of helicopters include: PID, Q-learning, model based and LQR control [4]–[7]. In [4], experimental results of a high performance adaptive augmentation technique applied to a LQR-based control law are presented. An adaptive swarm optimization based LQR control strategy is applied to a 2-DOF laboratory helicopter in [5]. A model-based control approach based on extended linearization techniques, utilizing unscented Kalman filters is presented in [6]. In [7], a multistep Q-learning (MsQL) method is developed based on the linearized model. Linear control strategies have the advantage of being relatively easy to understand and design. However, due to linearization of the model these techniques are effective only in a limited operating range.

The search for increased performance has always been the main goal for control engineers. As multirotor UAVs are highly nonlinear systems, simple control strategies may not be sufficient for the performance demand [8]. Therefore, our main focus is on nonlinear control strategies for 2-DOF helicopter systems. Some outstanding nonlinear control strategies in the field of helicopter systems are presented in [2], where neural backstepping and neural sliding mode block techniques are considered in order to control the pitch and yaw positions. The work in [1] presents a neuro-fuzzy system with a parameterized conjunctors and its performance is compared with a PID controller. An adaptive sliding mode observer based controller is designed and experimentally tested on 1-DOF (pitch) helicopter system in [9].

Due to the complexity of nonlinear feedback control, there is no systematic design procedure that applies to all nonlinear systems. Instead, a nonlinear control engineer must match a design procedure to a specific system with its objectives and specifications. Also, it is unlikely that the whole design of a nonlinear feedback controller can be based on one particular tool. Sliding mode control is one of these tools and is known to be a robust control technique. Moreover, sliding mode control has relatively good performance while having to deal with unmodeled effects, disturbances and uncertainties [10]. In order to meet the performance demand, sliding mode control is chosen as the primary control technique in this work.

Measured rotational positions and velocities suffer from measurement errors. Besides this, the measured velocities are subject to noise. To obtain more accurate velocity information, and overcome the dependency on sensors a sliding mode observer that estimates the angular velocities is proposed as part of our control strategy. Our work in [11] uses this control strategy to achieve asymptotic attitude regulation of a 3-DOF hover system to desired set points. In this paper, the control strategy is used to achieve asymptotic regulation of a 2-DOF helicopter system to desired set points as well as trajectory tracking. Rigorous Lyapunov-based analyses are provided to prove both the convergence of the helicopter to the desired set points, and the convergence of the estimated velocities to the actual velocities. Also, computer simulations and experimental results are presented to illustrate the effectiveness of the observer-based control law when applied to the complete

nonlinear system dynamics.

II. MATHEMATICAL MODEL

Consider the schematic model of a 2-DOF helicopter shown in Figure 1, where B denotes a right handed body-fixed frame with body-fixed axes x, y, z . The helicopter system

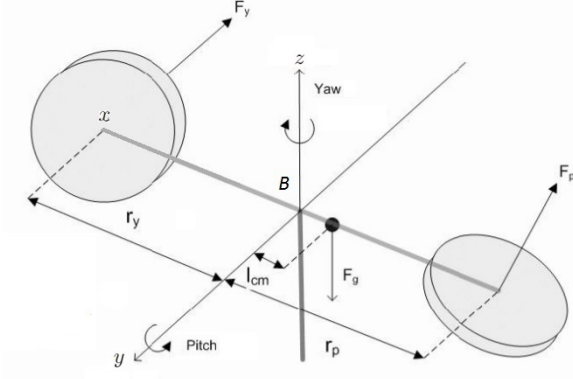


Figure 1. Schematic model of a 2-DOF helicopter system.

has two identical propellers that create thrust forces F_p and F_y at distances r_p and r_y , respectively, from the origin of the body-fixed frame. The first propeller (placed horizontally) generates a torque around the y -axis, which creates a pitch motion. The second propeller (placed vertically) generates a torque around the z -axis, which creates a yaw motion. The configuration variables are the pitch angle θ and the yaw angle ψ . The helicopter is a highly nonlinear multi-input-multi-output system, where the inputs are the voltages (V_p, V_y) to the DC-motors that control the propellers and the outputs are the pitch and yaw position of the system. The motor input voltages have a linear relation with generated torques. The system has mass m , and the center of gravity has its action point at distance l_{cm} from the origin of the body-fixed frame along the pitch axis. The moments of inertia of the rotating beam are J_p and J_y about the yaw and pitch motion, respectively. The constants of this problem are the gravity constant g , viscous friction coefficients D_p and D_y , and the thrust torque constants K_{pp}, K_{py}, K_{yp} , and K_{yy} .

The system is modeled using Lagrangian mechanics [2] which result in the following nonlinear equations of motion:

$$(J_p + ml_{cm}^2)\ddot{\theta} = K_{pp}V_p + K_{py}V_y - mgl_{cm}C(\theta) - D_p\dot{\theta} - ml_{cm}^2\dot{\psi}^2S(\theta)C(\theta), \quad (1)$$

$$(J_y + ml_{cm}^2C^2(\theta))\ddot{\psi} = K_{yp}V_p + K_{yy}V_y - D_y\dot{\psi} + 2ml_{cm}^2\dot{\psi}\dot{\theta}S(\theta)C(\theta), \quad (2)$$

where we have used the abbreviations $S(\cdot) = \sin(\cdot)$ and $C(\cdot) = \cos(\cdot)$.

To simplify the notation we introduce:

$$\begin{aligned} a_1 &= J_p + ml_{cm}^2, \\ a_2 &= -mgl_{cm}C(\theta) - D_p\dot{\theta} - ml_{cm}^2S(\theta)C(\theta)\dot{\psi}^2, \\ b_1 &= J_y + ml_{cm}^2C^2(\theta), \\ b_2 &= -D_y\dot{\psi} + 2ml_{cm}^2\dot{\psi}\dot{\theta}S(\theta)C(\theta). \end{aligned}$$

The input voltages are restricted to $V_p \in [-24, 24]$ V and $V_y \in [-24, 24]$ V.

III. SLIDING MODE CONTROL

This section presents a sliding mode control law that stabilizes the system (1), (2) to the desired set points (θ_d, ψ_d) . The control input is designed in a way that the trajectories of the system approach the sliding surface in finite time, and then stay on it for all future time. Thereto, the controller consists of an equivalent term which makes sure that the trajectories stay on the sliding surface for all future time, i.e. $\dot{s}_i = 0$, when $s_i = 0$, and a second term that makes sure the trajectory converges to the sliding surface. To realize this, the sliding surface is designed as $\dot{s}_i = -\lambda_i \text{sign}(s)$ where $\lambda_i > 0$, $i = \theta, \psi$.

Let $e_\theta = \theta^d - \theta$ and $e_\psi = \psi^d - \psi$ be the error variables. Define the following sliding variables:

$$\begin{aligned} s_\theta &= \dot{e}_\theta + k_\theta e_\theta, \\ s_\psi &= \dot{e}_\psi + k_\psi e_\psi, \end{aligned} \quad (3)$$

where $k_\theta > 0$ and $k_\psi > 0$, with time derivatives

$$\begin{aligned} \dot{s}_\theta &= \ddot{e}_\theta + k_\theta \dot{e}_\theta \\ &= -\frac{k_{pp}}{a_1}V_p - \frac{k_{py}}{a_1}V_y - \frac{a_2}{a_1} + k_\theta \dot{e}_\theta \\ &= -\lambda_\theta \text{sign}(s_\theta), \end{aligned} \quad (4)$$

and

$$\begin{aligned} \dot{s}_\psi &= \ddot{e}_\psi + k_\psi \dot{e}_\psi \\ &= -\frac{k_{yp}}{b_1}V_p - \frac{k_{yy}}{b_1}V_y - \frac{b_2}{b_1} + k_\psi \dot{e}_\psi \\ &= -\lambda_\psi \text{sign}(s_\psi), \end{aligned} \quad (5)$$

where sign is the signum function.

Using equations (4) and (5) the motor input voltages are extracted as

$$\begin{aligned} V_y &= \left(\frac{b_1 K_{pp}}{K_{pp}K_{yy} - K_{yp}K_{py}} \right) \left[\frac{K_{yp}a_2}{K_{pp}b_1} - \frac{K_{yp}a_1}{K_{pp}b_1}k_\theta \dot{e}_\theta \right. \\ &\quad \left. - \frac{K_{yp}a_1}{K_{pp}b_1}\lambda_\theta \text{sign}(s_\theta) - \frac{b_2}{b_1} + k_\psi \dot{e}_\psi + \lambda_\psi \text{sign}(s_\psi) \right], \\ V_p &= -\frac{K_{py}}{K_{pp}}V_y - \frac{a_2}{K_{pp}} + \frac{a_1}{K_{pp}}k_\theta \dot{e}_\theta + \frac{a_1}{K_{pp}}\lambda_\theta \text{sign}(s_\theta). \end{aligned}$$

The time derivative of the Lyapunov function

$$V(s_\theta, s_\psi) = \frac{1}{2}(s_\theta^2 + s_\psi^2) \quad (6)$$

can be computed as

$$\dot{V}(s_\theta, s_\psi) = -\lambda_\theta |s_\theta| - \lambda_\psi |s_\psi| < 0, \quad (7)$$

which shows that the system approaches the sliding surface in finite time and stays on the sliding surface for all future time. The dynamics on the sliding surface are

$$\dot{e}_i = -k_i e_i, \quad (8)$$

$i = \theta, \psi$, which implies that e_i converge to zero exponentially fast.

IV. OBSERVER DESIGN

The system is equipped with sensors to measure the angular velocities. However, the measured velocities are subject to noise. To obtain more accurate rate information and overcome the sensor dependency, we propose a sliding mode observer that estimates the angular velocities. Let $x_1 = [\theta, \psi]^T$ and $x_2 = \dot{x}_1$. Then, the model (1), (2) can be written as

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= f(x_2) + g(x_1)U, \end{aligned} \quad (9)$$

where $U = [V_p, V_y]^T$ and

$$f(x_2) = \begin{bmatrix} \frac{a_2}{b_1} \\ \frac{a_1}{b_2} \end{bmatrix}, \quad g(x_1) = \begin{bmatrix} \frac{K_{pp}}{K_{yp}} & \frac{K_{py}}{K_{yy}} \\ \frac{a_1}{b_1} & \frac{a_1}{b_1} \end{bmatrix}.$$

Let $\hat{x} = [\hat{x}_1^T, \hat{x}_2^T]^T$ denote the estimated state vector and consider the following observer

$$\begin{aligned} \dot{\hat{x}}_1 &= \hat{x}_2 + \underline{K} \text{sign}(x_1 - \hat{x}_1), \\ \dot{\hat{x}}_2 &= f(\hat{x}_2) + g(\hat{x}_1)U + \underline{L} \underline{K} \text{sign}(x_1 - \hat{x}_1), \end{aligned} \quad (10)$$

where $\underline{K} = \text{diag}\{\kappa_1, \kappa_2\}$ and $\underline{L} = \text{diag}\{l_1, l_2\}$ are the observer gain matrices. Denote by $\tilde{x} = x - \hat{x}$ the estimation error. Then, the estimation error dynamics are expressed as

$$\begin{aligned} \dot{\tilde{x}}_1 &= \tilde{x}_2 - \underline{K} \text{sign}(\tilde{x}_1), \\ \dot{\tilde{x}}_2 &= f(\tilde{x}_2) - f(\hat{x}_2) + (g(x_1) - g(\hat{x}_1))U - \underline{L} \underline{K} \text{sign}(\tilde{x}_1). \end{aligned} \quad (11)$$

First define $c_2 = \frac{D_p}{a_1}$, $c_3 = \frac{ml_{cm}^2}{a_1}$, $c_4 = \frac{D_y}{b_1}$, and $c_5 = \frac{2ml_{cm}^2}{b_1}$, and then we assume that $c_2 \approx c_3 \approx c_4 \approx c_5 := c$, and $a_1 \approx b_1 := a$. The following result can now be stated.

Proposition 2 Consider the observer (10) and assume that positive gains κ_i , l_i , $i = 1, 2$ satisfy

$$\kappa_1 > |\dot{\theta}|, \quad \kappa_2 > |\dot{\psi}|, \quad (12a)$$

$$l_1 > \frac{c}{a}, \quad (12b)$$

$$(l_1 - \frac{c}{a})(l_2 - \frac{c}{a} + \zeta_1) > \zeta_2, \quad (12c)$$

where $\zeta_1 = \dot{\theta}^2 \frac{c}{a} C(\theta)S(\theta)$, and $\zeta_2 = \dot{\psi}^2 \frac{c^2}{a^2} C^2(\theta)S^2(\theta)$. Then, the estimation error $\tilde{x}$ converges to zero.

Proof: Consider the observer error dynamics (11). The condition (12a) guarantees that the sliding manifold $\tilde{x}_1 = 0$ is reached in finite time. During sliding $\dot{\tilde{x}}_1 \approx 0$ and thus $\tilde{x}_2 = \underline{K} \text{sign}(\tilde{x}_1)$. So

$$\begin{aligned} \dot{\tilde{x}}_2 &= -\underline{L} \tilde{x}_2 + \begin{bmatrix} \frac{c}{a} & 0 \\ 0 & \frac{c}{a} \end{bmatrix} \tilde{x}_2 \\ &+ C(\theta)S(\theta) \begin{bmatrix} \frac{c}{a}(\dot{\psi}^2 - \dot{\psi}^2) \\ -\frac{c}{a}(\dot{\theta}\dot{\psi} - \dot{\theta}\dot{\psi}) \end{bmatrix} + \Delta_g U. \end{aligned} \quad (13)$$

Because Δ_g only depends on x_1 , $\Delta_g \approx 0$.

Consider the Lyapunov function candidate

$$V(\tilde{x}_2) = \frac{1}{2}(\dot{\theta}^2 + \dot{\psi}^2). \quad (14)$$

Taking the time derivative along the trajectories of (13) yields

$$\begin{aligned} \dot{V} &= -(l_1 - \frac{c}{a})\dot{\theta}^2 - (l_2 - \frac{c}{a})\dot{\psi}^2 + C(\theta)S(\theta)\frac{c}{a}(\dot{\psi}\dot{\theta}\dot{\psi} - \dot{\psi}^2\dot{\theta}) \\ &= -\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix}^T \begin{bmatrix} l_1 - \frac{c}{a} & \frac{c}{2b}\dot{\psi}^2 C(\theta)S(\theta) \\ \frac{c}{2b}\dot{\psi}^2 C(\theta)S(\theta) & l_2 - \frac{c}{a} + \frac{c}{a}\dot{\theta}C(\theta)S(\theta) \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix}. \end{aligned}$$

The conditions (12b) and (12c) imply that all the principle minors of (IV) are positive definite and therefore $\dot{V}(\tilde{x}_1, \tilde{x}_2)$ is negative definite. Thus, the estimation error converges to zero at an exponential rate. Hence, the states $\tilde{x}_2$ are obtained exponentially fast. The observer gains l_i are chosen relatively large so that the observer converges much faster than the system dynamics.

V. SIMULATIONS

This section illustrates the previously developed ideas via numerical simulations using the system parameters for Quanser AERO 2-DOF helicopter, which are given in Table I.

Table I
SYSTEM PARAMETERS

	Parameter	Value	Unit
J_p	Moment of inertia about the pitch axis	0.0215	kg · m ²
J_y	Moment of inertia about the yaw axis	0.0237	kg · m ²
l_{cm}	Center of mass distance from the body-fixed frame origin	0.0071	m
m	Mass	1.0750	kg
D_p	Pitch viscous friction constant	0.0071	N/V
D_y	Yaw viscous friction constant	0.0220	N/V
K_{pp}	Torque thrust gain acting on pitch axis from pitch propeller	0.0011	N · m/V
K_{py}	Torque thrust gain acting on pitch axis from yaw propeller	0.0021	N · m/V
K_{yp}	Torque thrust gain acting on yaw axis from pitch propeller	-0.0027	N · m/V
K_{yy}	Torque thrust gain acting on yaw axis from yaw propeller	0.0022	N · m/V

Remark 1: In the simulations and experiments, to alleviate chattering and speed up the simulations, a smooth approximation of the discontinuous signum function is used:

$$\text{sign}(s) \approx \tanh(\alpha s), \quad (15)$$

where α is a sufficiently large positive constant. The stability properties of the system with the smooth implementation of the controllers can be analyzed. Following [12], it can be shown that the closed-loop system is globally uniformly ultimately bounded with respect to a compact set around the origin. Therefore, the system converges to a small neighborhood of the origin. This set can be made arbitrarily small by increasing the slope α of the function $\tanh(\alpha s)$.

The initial conditions for the simulations are:

$$\begin{aligned} (\theta_0, \psi_0) &= (0, 0), (\hat{\theta}_0, \hat{\psi}_0) = (0, 0), \\ (\dot{\theta}_0, \dot{\psi}_0) &= (0.5, 0.7) \text{ [rad/s]}, (\hat{\dot{\theta}}_0, \hat{\dot{\psi}}_0) = (0, 0). \end{aligned}$$

The desired positions are chosen as $\theta_d = 1$ rad, $\psi_d = 2$ rad and the control parameters are:

$$\begin{aligned} \lambda_\theta &= 12, \lambda_\psi = 5, \\ k_\theta &= 5, k_\psi = 5, \\ l_1 &= 40, l_2 = 40, \\ \kappa_1 &= 2, \kappa_2 = 2, \\ \alpha &= 3. \end{aligned}$$

Figure 2 shows that the estimated velocities converge to the actual velocities in less than one second. Therefore, the observer-based control law achieves the same convergence to the desired set points as the control law based on the actual states. As shown in Figure 3 the control law developed in the previous sections makes the angles converge to the desired set points.

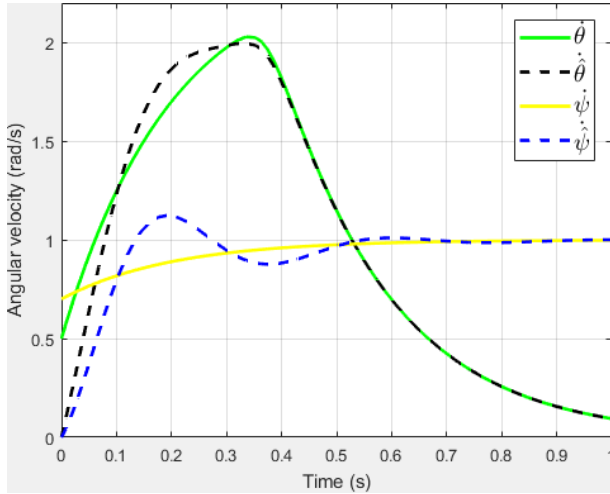


Figure 2. Time response of the angular velocities.

VI. EXPERIMENTAL RESULTS

This section illustrates the experimental implementation of the control strategy on the Quanser Aero helicopter system. The Quanser Aero helicopter system has a default PID-controller, this PID-controller is used as comparison to the proposed observer-based sliding mode control (SMC) strategy.

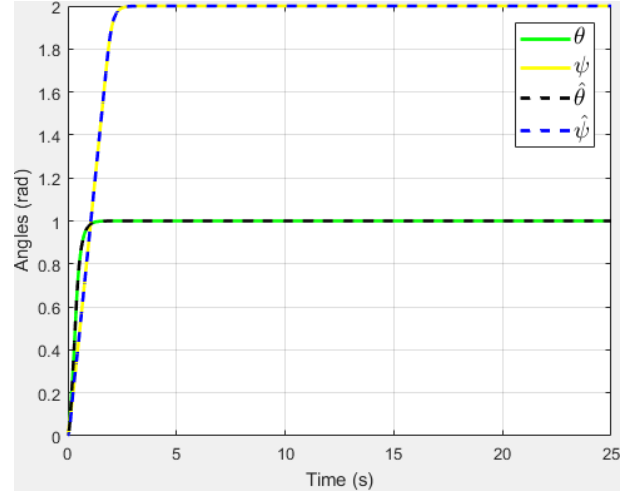


Figure 3. Time response of the angular positions.

All the system and control parameters are the same as in the simulations.

The initial conditions for the experiments are:

$$\begin{aligned} (\theta_0, \psi_0) &= (-45, -10) \text{ [deg]}, \\ (\hat{\theta}_0, \hat{\psi}_0) &= (0, 0), (\dot{\hat{\theta}}_0, \dot{\hat{\psi}}_0) = (0, 0). \end{aligned}$$

For the pitch angle the desired trajectory is a square wave signal with an amplitude of 10 degrees and a frequency of 0.04 Hz. For the yaw angle the desired trajectory is a square wave signal with an amplitude of 45 degrees and a frequency of 0.04 Hz.

In Figures 4 and 5 the time response of the observer based SMC control strategy is compared to the PID-controller. Figure 4 shows the pitch angle and Figure 5 shows the yaw angle. Both plots show a significant improvement of the observer-based SMC law compared to the PID-controller. The SMC strategy reduces the steady state error with 64% in the pitch dynamics, and 81.5% in the yaw dynamics. Figure 6 shows the input voltages to the pitch and yaw motors, as can be seen, are within the allowable limits, i.e., $V_p \in [-24, 24]$ V and $V_y \in [-24, 24]$ V.

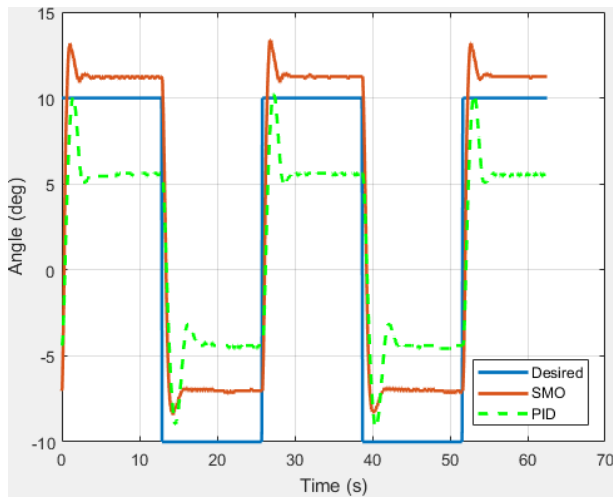


Figure 4. Time response of the pitch angle.

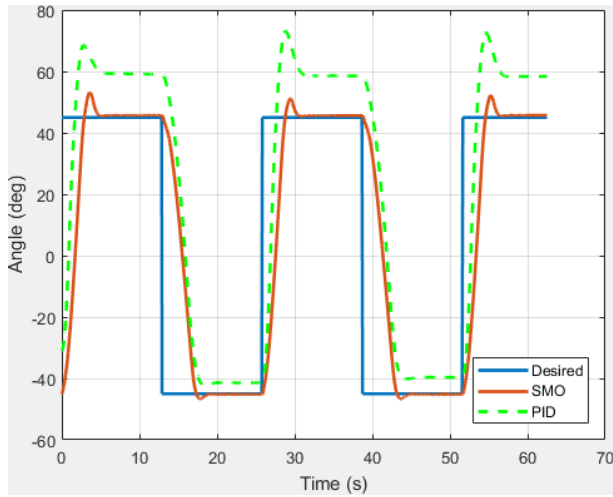


Figure 5. Time response of the yaw angle.

VII. CONCLUSIONS

An observer-based sliding mode control strategy is presented for a 2-DOF helicopter. Rigorous Lyapunov-based analyses are provided to prove both the convergence of the helicopter to the desired set points, and the convergence of the estimated velocities to the actual velocities. Numerical simulations are provided to illustrate the capability of the control law to achieve convergence to the desired set points when applied to the complete nonlinear system dynamics. Ultimately, the control strategy is experimentally implemented on the Quanser Aero 2-DOF helicopter and the results are compared to a conventional PID-controller.

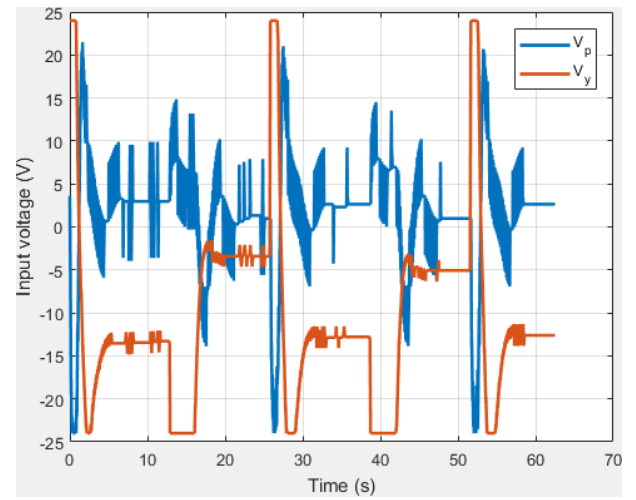


Figure 6. Motor input voltages.

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REFERENCES

- [1] A.C. Aras, and O. Kaynak, "Trajectory tracking of a 2-DOF Helicopter System using Neuro-Fuzzy System with Parameterized Conjunctors," *International Conference on Advanced Intelligent Mechatronics*, 2014.
- [2] M. Hernandez-Gonzalez, A.Y. Alanis, and E.A. Hernandez-Vargas "Decentralized discrete-time neural control for a Quanser 2-DOF helicopter," *Applied soft computing*, 2012, Vol. 12, pp. 2462-2469.
- [3] Quanser Inc., Quanser AERO Laboratory Guide, *Tech. rep.*, Quanser 2016.
- [4] P. Nuthi, and K. Subbarao, "Experimental Verification of Linear and Adaptive Control Techniques for a Two Degrees-of-Freedom Helicopter," *Journal of Dynamic Systems, Measurement, and Control*, 2015, Vol. 137.
- [5] E. V. Kumar, G.S.Raaja, J. Jerome, "Adaptive PSO for optimal LQR tracking control of 2 DoF laboratory helicopter," *Applied Soft Computing*, 2016, Vol. 41, pp. 77-90.
- [6] S. S. Butt, H. Sun, and H. Aschemann, "Control Design by extended linearisation techniques for a two degrees of freedom helicopter," *International Federation of Automatic Control Papers Online*, 2015, Vol. 48-11, pp. 22-27.
- [7] B. Luo, H.-N. Wu, T. Huang, "Optimal Output Regulation for Model-Free Quanser Helicopter With Multistep Q-Learning," *IEEE Transaction on Industrial Electronics*, VOL. 65, NO. 6, JUNE 2018.
- [8] J. Ligthart, P. Poksawat, L. Wang, and H. Nijmeijer, "Experimentally validated model predictive control for a hexacopter," *IFAC PapersOnLine*, 2017, pp. 4076-4081.
- [9] K. Khayati, "Multivariable Adaptive Sliding-Mode Observer-Based Control for Mechanical Systems," *Canadian Journal of Electrical, and Computer Engineering*, 2015, Vol. 38, No. 3.
- [10] H. K. Khalil, *Nonlinear systems*, Prentice Hall, New Jersey, USA, 2002.
- [11] M. Reyhanoglu, R. Damen, and W. MacKunis, "Observer-based sliding mode control of a 3-DOF hover system," *Proceedings of International Conference on Control, Automation, Robotics and Vision*, 2016, pp. 1-6.
- [12] F. Esfandiari and H. K. Khalil, "Stability analysis of a continuous implementation of variable structure control," *IEEE Transactions on Automatic Control*, Vol. 36, No. 5, 1991, pp. 616-620.